

RESEARCH INTEREST

- **Statistical Learning, Machine Learning (DL, RL, LLM), Optimization Theory, Financial Engineering**

EDUCATION

Peking University

Beijing, China

- *B.A. in Mathematics, School of Mathematical Sciences*

Jun. 2024–Jul. 2028 (Expected)

Second Major B.A. in Economics, National School of Development

Sep. 2025–Jul. 2028 (Expected)

- **Mathematics Courses:** Mathematical Analysis, Advanced Algebra, Geometry, Mathematical Statistics, Statistical Thinking, Stochastic Analysis and Applications.
- **Mathematics Courses (Current Semester):** Probability Theory, Complex Analysis, Ordinary Differential Equations, Applied Stochastic Processes.
- **Computer Science Courses:** Machine Learning and Artificial Intelligence, Fundamentals of Artificial Intelligence
- **Finance and Economics Courses:** Principles of Economics, Intermediate Microeconomics, Introduction to Finance, Security Investment, Mathematical Methods in Finance, Modern Quantitative Trading Systems, Topics in Quantitative Finance

University of Copenhagen

Copenhagen, Denmark

- *International Exchange Student, Department of SCIENCE*

Sep. 2025–Jan. 2026

- **Mathematics Courses:** Topics in Statistics (MCMC), Advanced Operation Research: Stochastic Programming.
- **Computer Science Courses:** Probability Machine Learning

Peking University

Beijing, China

- *B.A. in Physics, School of Earth and Space Sciences*

Sep. 2023–Jun. 2024

RESEARCH EXPERIENCE ON STATISTICAL LEARNING AND MACHINE LEARNING

Optimization on Tensor Regression Model

HKUST

- *Advisor: Prof. Yi Chen, HKUST & Prof. Biao Cai, CityU*

Jul. 2025 – Present

- **Algorithm Implementation for Partially Observed Dynamic Tensor Regression**

Ongoing Research Project

- * Investigated and implemented the POSTER framework for ultra-high-dimensional dynamic tensor regression tasks, specifically handling complex random and block missing data patterns.
- * Initiated theoretical convergence analysis of the non-convex alternating minimization framework, utilizing tools from High-Dimensional Probability to try to establish statistical guarantees and error bounds under sparsity and structural constraints.
- * Designed robust simulation pipelines to evaluate tensor structure recovery.

Applications of LLM Agent

Peking University

- *Advisor: Prof. Shanghang Zhang, Peking University*

Apr. 2025 – Jun. 2025

- **Wechat Agent**

Course Project

GitHub

- * Developed an LLM-driven typesetting plugin on the Dify platform, automating the end-to-end content generation and draft publishing pipeline via Python and WeChat APIs.
- * Engineered structural prompts using DeepSeek-VL v3 to enforce strict HTML and CSS constraints, ensuring the zero-shot generation of platform-compliant, mobile-responsive layouts.
- * Integrated a RAG-based writing assistant using embedding models and orchestrated external APIs (e.g., Firecrawl, Bing Search) for dynamic web scraping, data retrieval, and content compliance.
- * Evaluated the impact of generation parameters on layout stability and authored comprehensive technical documentation detailing the system architecture and experimental results.

RESEARCH EXPERIENCE ON FINANCE

Modern Quantitative Trading Systems

Peking University

Sep. 2024 – Jan. 2025

▪ Advisor: Prof. Changhao Jiang, Peking University

◦ Quantitative Trading Backtesting Framework

[GitHub](#)

Course Project

- * Engineered a modular, object-oriented quantitative backtesting framework in Python, decoupling signal generation from order execution to support highly customizable trading algorithms.
- * Designed and implemented diverse quantitative strategies, including Momentum, Mean Reversion (Bollinger Bands), Cross-Sectional, and Time-Series, utilizing pandas and NumPy for vectorized data processing.
- * Developed an analytics module to compute key performance indicators (e.g., Sharpe Ratio, Annualized Returns) and automated the generation of visual performance reports for systematic strategy evaluation.

Security Investment

Peking University

Mar. 2025 – Jun. 2025

▪ Advisor: Prof. Hai Huang, Peking University

◦ Empirical Asset Pricing and Portfolio Management

[GitHub](#)

Course Project

- * Conducted empirical research on the Chinese A-share market, analyzing risk-return profiles, market volatility drivers, and sector-specific performance across historical market cycles.
- * Constructed optimal investment portfolios utilizing the Markowitz Mean-Variance framework, integrating fundamental quantitative stock selection and backtesting strategies across dynamic rebalancing frequencies.
- * Evaluated mutual fund performance by implementing Treynor-Mazuy (TM) and Henriksson-Merton (HM) models to quantitatively assess fund managers' market-timing and stock-picking capabilities.

SKILLS & OTHERS

- **Computer Skills:** Python (&PyTorch), R, C++ (Elementary), LaTeX
- **Language:** Mandarin Chinese (Native), English (Fluent)
- **The Graduate Record Examinations (GRE):** Total 325 with 155 (Verbal) + 170 (Quant)